

Physics Questions - motion in 1 d include graphs also

Subject: Physics | Chapter: motion in 1 d include graphs also | Difficulty: hard

Question 1: A particle moves along the x-axis with its velocity given by $v(t) = 6t^2 - 12t + 6$ m/s. At $t = 0$, its position is $x = 2$ m. What is its position at $t = 2$ s? See graph for visualization. (Graph would show a parabola representing $v(t)$)

Options:

- A) 4 m
- B) 6 m
- C) 8 m
- D) 10 m

Difficulty: 8/5

Question 2: A ball is thrown vertically upwards with an initial velocity of 20 m/s. Ignoring air resistance, at what time(s) will its speed be 10 m/s? ($g = 10 \text{ m/s}^2$). See graph for visualization (Graph would show a parabola representing velocity vs time)

Options:

- A) 1s only
- B) 1s and 3s
- C) 2s only
- D) 3s only

Difficulty: 7/5

Question 3: A car accelerates uniformly from rest to 20 m/s in 5 seconds. Then it maintains this speed for 10 seconds before decelerating uniformly to rest in 2 seconds. Calculate the total distance traveled. See graph for visualization (Graph would show a trapezoid representing velocity vs time)

Options:

- A) 200 m
- B) 210 m
- C) 220 m
- D) 230 m

Difficulty: 9/5

Question 4: Two objects are dropped from the same height, one twice as heavy as the other. Ignoring air resistance, which statement is true? (Graph would show two overlapping lines representing position vs time for both objects)

Options:

- A) The heavier object hits the ground first.
- B) The lighter object hits the ground first.
- C) They hit the ground at the same time.
- D) Their speeds are different throughout the fall.

Difficulty: 6/5

Question 5: A projectile is launched at an angle of 30° with an initial velocity of 40 m/s. What is the horizontal range of the projectile ($g = 10 \text{ m/s}^2$)? (Graph would show a parabolic trajectory)

Options:

- A) 80" 3 m
- B) 160" 3 m
- C) 80 m
- D) 160 m

Difficulty: 8/5

Question 6: A body is moving with a velocity of 10 m/s. It experiences a constant deceleration of 2 m/s^2 . How far does it travel before coming to rest? (Graph would show a triangle representing velocity vs time)

Options:

- A) 20 m
- B) 25 m
- C) 50 m
- D) 100 m

Difficulty: 7/5

Question 7: A particle's position is given by $x(t) = t^3 - 6t^2 + 9t$. At what time(s) is the particle at rest? (Graph would show a cubic curve representing position vs time)

Options:

- A) $t = 1\text{s}$ and $t = 3\text{s}$
- B) $t = 0\text{s}$ and $t = 3\text{s}$
- C) $t = 1\text{s}$ only
- D) $t = 3\text{s}$ only

Difficulty: 9/5

Question 8: A train travels 120 km at 60 km/h and then 120 km at 40 km/h. What is the average speed for the entire journey?

Options:

- A) 48 km/h
- B) 50 km/h
- C) 52 km/h
- D) 55 km/h

Difficulty: 7/5

Question 9: A freely falling body travels 45m in the last second of its fall. What is the height from which it was dropped? ($g = 10 \text{ m/s}^2$)

Options:

- A) 125 m
- B) 100 m
- C) 80 m
- D) 50 m

Difficulty: 10/5

Question 10: A rocket is launched vertically upwards with a constant acceleration of 20 m/s^2 . After 10 seconds, its engine shuts off. What is the maximum height it reaches? ($g = 10 \text{ m/s}^2$)

Options:

- A) 2000 m
- B) 2500 m
- C) 3000 m
- D) 3500 m

Difficulty: 9/5

Generated on: 30/7/2025, 3:26:38 am